Group	PD	HC	Post-Hoc
Subjects	423 (61.9%)	196 (28.7%)	
Female	146 (34.5%)	70 (35.7%)	None
APOE4	101 (26.3%) n=384	47 (26.4%) n=178	None
Age	61.7 (9.7) (33.5, 84.9)	60.8 (11.2) (30.6, 83.7)	None
Education	15.56 (2.97) (5.00, 26.00)	16.04 (2.89) (8.00, 24.00)	None
GDS	2.32 (2.44) (0.00, 14.00)	1.29 (2.10) (0.00, 15.00)	a
UPSIT	22.35 (8.23) (1.00, 40.00)	33.98 (4.86) (11.00, 40.00)	a
REM	4.12 (2.69) (0.00, 12.00)	2.83 (2.26) (0.00, 11.00)	a
UPDRS3	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
UPDRSTOT	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRSTOTOM	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRS1	5.57 (4.06) (0.00, 24.00)	2.95 (2.96) (0.00, 17.00)	a
UPDRS2	5.90 (4.19) (0.00, 22.00)	0.46 (1.02) (0.00, 6.00)	a
UPDRS3ON	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
MOCA	27.13 (2.32) (17.00, 30.00)	28.23 (1.11) (26.00, 30.00)	a
LNS	10.59 (2.66) (2.00, 20.00)	10.87 (2.57) (2.00, 20.00)	None
SDMT	41.18 (9.73) (7.00, 82.00)	46.77 (10.53) (20.00, 83.00)	a
JLO	12.77 (2.13) (5.00, 15.00)	13.12 (1.98) (4.00, 15.00)	None
INMEDIATE	24.44 (4.98) (9.00, 36.00)	26.05 (4.50) (15.00, 35.00)	a
HVLTRDLY	8.36 (2.52) (0.00, 12.00)	9.29 (2.32) (2.00, 12.00)	a
SFT	48.67 (11.63) (20.00, 103.00)	51.80 (11.20) (22.00, 80.00)	a
ADL	93.14 (5.89) (70.00, 100.00)	100.00 (nan) (100.00, 100.00)	None
SCOPA	9.51 (6.18) (0.00, 39.00)	5.83 (3.69) (0.00, 20.00)	a
ABETA	909.55 (410.73) (238.80, 3707.00)	1019.37 (499.46) (239.10, 3297.00)	a
ASYN	1506.70 (666.59) (432.40, 5256.90)	1695.19 (747.42) (488.60, 4683.10)	a
TAU	169.54 (56.80) (80.93, 467.00)	191.64 (79.26) (81.96, 580.80)	a
PTAU	14.87 (5.25) (8.01, 40.13)	17.52 (8.35) (8.08, 73.61)	a
pTauAB	0.02 (0.01) (0.01, 0.08)	0.02 (0.02) (0.01, 0.18)	None
MEANPUT	118.11 (366.15) (0.24, 1775.00)	1163.34 (1184.42) (0.64, 3695.00)	a
MEANCAU	1084.02 (1088.86) (0.39, 3705.00)	1639.12 (1563.49) (1.32, 5195.00)	a

Group	PD	HC	Post-Hoc
Subjects	423 (61.9%)	196 (28.7%)	
Female	146 (34.5%)	70 (35.7%)	None
APOE4	101 (26.3%) n=384	47 (26.4%) n=178	None
Age	61.7 (9.7) (33.5, 84.9)	60.8 (11.2) (30.6, 83.7)	None
Education	15.56 (2.97) (5.00, 26.00)	16.04 (2.89) (8.00, 24.00)	None
GDS	2.32 (2.44) (0.00, 14.00)	1.29 (2.10) (0.00, 15.00)	a
UPSIT	22.35 (8.23) (1.00, 40.00)	33.98 (4.86) (11.00, 40.00)	a
REM	4.12 (2.69) (0.00, 12.00)	2.83 (2.26) (0.00, 11.00)	a
UPDRS3	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
UPDRSTOT	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRSTOTOM	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRS1	5.57 (4.06) (0.00, 24.00)	2.95 (2.96) (0.00, 17.00)	a
UPDRS2	5.90 (4.19) (0.00, 22.00)	0.46 (1.02) (0.00, 6.00)	a
UPDRS3ON	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
MOCA	27.13 (2.32) (17.00, 30.00)	28.23 (1.11) (26.00, 30.00)	a
LNS	10.59 (2.66) (2.00, 20.00)	10.87 (2.57) (2.00, 20.00)	None
SDMT	41.18 (9.73) (7.00, 82.00)	46.77 (10.53) (20.00, 83.00)	a
JLO	12.77 (2.13) (5.00, 15.00)	13.12 (1.98) (4.00, 15.00)	None
INMEDIATE	24.44 (4.98) (9.00, 36.00)	26.05 (4.50) (15.00, 35.00)	a
HVLTRDLY	8.36 (2.52) (0.00, 12.00)	9.29 (2.32) (2.00, 12.00)	a
SFT	48.67 (11.63) (20.00, 103.00)	51.80 (11.20) (22.00, 80.00)	a
ADL	93.14 (5.89) (70.00, 100.00)	100.00 (nan) (100.00, 100.00)	None
SCOPA	9.51 (6.18) (0.00, 39.00)	5.83 (3.69) (0.00, 20.00)	a
ABETA	909.55 (410.73) (238.80, 3707.00)	1019.37 (499.46) (239.10, 3297.00)	a
ASYN	1506.70 (666.59) (432.40, 5256.90)	1695.19 (747.42) (488.60, 4683.10)	a
TAU	169.54 (56.80) (80.93, 467.00)	191.64 (79.26) (81.96, 580.80)	a
PTAU	14.87 (5.25) (8.01, 40.13)	17.52 (8.35) (8.08, 73.61)	a
pTauAB	0.02 (0.01) (0.01, 0.08)	0.02 (0.02) (0.01, 0.18)	None
MEANPUT	0.83 (0.30) (0.24, 2.17)	2.15 (0.56) (0.64, 3.89)	a
MEANCAU	2.00 (0.56) (0.39, 3.71)	2.98 (0.63) (1.32, 5.20)	a

Group	PD	HC	Post-Hoc
Subjects	423 (61.9%)	196 (28.7%)	
Female	146 (34.5%)	70 (35.7%)	NaN
APOE4	101 (26.3%) n=384	47 (26.4%) n=178	NaN
Age	61.7 (9.7) (33.5, 84.9)	60.8 (11.2) (30.6, 83.7)	NaN
Education	15.56 (2.97) (5.00, 26.00)	16.04 (2.89) (8.00, 24.00)	NaN
GDS	2.32 (2.44) (0.00, 14.00)	1.29 (2.10) (0.00, 15.00)	a
UPSIT	22.35 (8.23) (1.00, 40.00)	33.98 (4.86) (11.00, 40.00)	a
REM	4.12 (2.69) (0.00, 12.00)	2.83 (2.26) (0.00, 11.00)	a
UPDRS3	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
UPDRSTOT	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRSTOTOM	32.35 (13.13) (7.00, 72.00)	4.56 (4.40) (0.00, 20.00)	a
UPDRS1	5.57 (4.06) (0.00, 24.00)	2.95 (2.96) (0.00, 17.00)	a
UPDRS2	5.90 (4.19) (0.00, 22.00)	0.46 (1.02) (0.00, 6.00)	a
UPDRS3ON	20.89 (8.85) (4.00, 51.00)	1.21 (2.19) (0.00, 13.00)	a
MOCA	27.13 (2.32) (17.00, 30.00)	28.23 (1.11) (26.00, 30.00)	a
LNS	10.59 (2.66) (2.00, 20.00)	10.87 (2.57) (2.00, 20.00)	NaN
SDMT	41.18 (9.73) (7.00, 82.00)	46.77 (10.53) (20.00, 83.00)	a
JLO	12.77 (2.13) (5.00, 15.00)	13.12 (1.98) (4.00, 15.00)	NaN
INMEDIATE	24.44 (4.98) (9.00, 36.00)	26.05 (4.50) (15.00, 35.00)	a
HVLTRDLY	8.36 (2.52) (0.00, 12.00)	9.29 (2.32) (2.00, 12.00)	a
SFT	48.67 (11.63) (20.00, 103.00)	51.80 (11.20) (22.00, 80.00)	a
ADL	93.14 (5.89) (70.00, 100.00)	100.00 (nan) (100.00, 100.00)	NaN
SCOPA	9.51 (6.18) (0.00, 39.00)	5.83 (3.69) (0.00, 20.00)	a
ABETA	909.55 (410.73) (238.80, 3707.00)	1019.37 (499.46) (239.10, 3297.00)	a
ASYN	1506.70 (666.59) (432.40, 5256.90)	1695.19 (747.42) (488.60, 4683.10)	a
TAU	169.54 (56.80) (80.93, 467.00)	191.64 (79.26) (81.96, 580.80)	a
PTAU	14.87 (5.25) (8.01, 40.13)	17.52 (8.35) (8.08, 73.61)	a
pTauAB	0.02 (0.01) (0.01, 0.08)	0.02 (0.02) (0.01, 0.18)	NaN
MEANPUT	0.83 (0.30) (0.24, 2.17)	2.15 (0.56) (0.64, 3.89)	a
MEANCAU	2.00 (0.56) (0.39, 3.71)	2.98 (0.63) (1.32, 5.20)	a

Group	PD_CU	PD_MCI	PD_DEMENTIA	p-value	Post-Hoc
Subjects	323 (47.3%)	83 (12.2%)	17 (2.5%)		
Female	120 (37.2%)	20 (24.1%)	6 (35.3%)		None
APOE4	78 (26.4%) n=295	18 (23.7%) n=76	5 (38.5%) n=13		None
Age	60.1 (9.8) (33.5, 82.3)	66.0 (7.4) (48.0, 84.9)	69.4 (7.4) (54.8, 83.0)		a,b
Education	15.70 (2.87) (8.00, 26.00)	14.99 (2.82) (8.00, 21.00)	15.59 (4.91) (5.00, 21.00)		None
GDS	1.07 (1.59) (0.00, 10.00)	1.63 (1.69) (0.00, 10.00)	1.71 (1.45) (0.00, 5.00)		
UPSIT	23.41 (7.87) (5.00, 39.00)	19.54 (8.60) (1.00, 40.00)	16.06 (7.39) (5.00, 31.00)		a,b
REM	2.61 (2.23) (0.00, 12.00)	3.22 (2.30) (0.00, 11.00)	3.71 (3.24) (0.00, 11.00)		None
UPDRS3	18.30 (8.49) (2.00, 46.00)	20.25 (8.94) (4.00, 51.00)	19.35 (8.26) (9.00, 39.00)		None
UPDRSTOT	28.81 (12.37) (5.00, 72.00)	33.23 (12.78) (7.00, 68.00)	35.41 (13.18) (18.00, 70.00)		
UPDRSTOTOM	26.61 (12.04) (3.00, 72.00)	31.07 (12.21) (7.00, 60.00)	33.24 (14.17) (12.00, 63.00)		a
UPDRS1	3.97 (3.39) (0.00, 19.00)	5.48 (4.01) (0.00, 18.00)	6.88 (3.48) (2.00, 15.00)		a,b
UPDRS2	4.33 (3.70) (0.00, 22.00)	5.31 (3.98) (0.00, 18.00)	6.88 (3.57) (2.00, 13.00)		
UPDRS3ON	15.42 (8.33) (0.00, 42.00)	17.30 (8.17) (1.00, 42.00)	16.65 (9.12) (4.00, 33.00)		None
MOCA	25.47 (2.57) (15.00, 30.00)	22.51 (3.62) (9.00, 29.00)	16.53 (5.51) (2.00, 23.00)		a,b,c
LNS	9.12 (2.40) (1.00, 16.00)	7.00 (2.64) (2.00, 14.00)	3.71 (3.24) (0.00, 10.00)		a,b,c
SDMT	36.70 (8.74) (7.00, 59.00)	27.22 (11.09) (2.00, 54.00)	13.41 (13.87) (0.00, 48.00)		a,b,c
JLO	11.17 (2.24) (2.00, 15.00)	9.84 (2.85) (0.00, 15.00)	8.06 (3.13) (2.00, 13.00)		a,b
INMEDIATE	21.56 (4.72) (8.00, 32.00)	17.08 (5.07) (4.00, 31.00)	10.41 (4.15) (4.00, 19.00)		a,b,c
HVLTRDLY	6.79 (2.88) (0.00, 12.00)	4.69 (3.06) (0.00, 11.00)	1.53 (2.24) (0.00, 6.00)		a,b,c
SFT	44.34 (9.64) (23.00, 74.00)	36.20 (9.23) (16.00, 58.00)	23.06 (11.00) (7.00, 45.00)		a,b,c
ADL	84.47 (9.15) (20.00, 100.00)	78.49 (10.32) (50.00, 95.00)	57.06 (23.59) (20.00, 90.00)		a,b,c
SCOPA	7.11 (4.84) (0.00, 35.00)	9.83 (5.79) (0.00, 32.00)	10.94 (5.86) (4.00, 23.00)		a,b
ABETA	800.21 (312.32) (267.40, 2260.00)	740.75 (363.57) (238.80, 2396.00)	525.86 (209.41) (271.50, 952.60)		b
ASYN	1292.33 (558.13) (336.10, 4223.30)	1273.67 (540.39) (458.80, 2766.60)	1217.69 (580.52) (356.10, 2226.50)		None
TAU	153.28 (50.44) (80.88, 349.80)	163.30 (55.69) (81.11, 328.10)	182.53 (77.31) (80.98, 327.40)		None
PTAU	13.42 (4.57) (8.02, 31.64)	14.41 (5.16) (8.09, 28.43)	17.31 (7.23) (8.01, 30.84)		b
pTauAB	0.01 (0.01) (0.01, 0.06)	0.02 (0.01) (0.01, 0.08)	0.03 (0.02) (0.01, 0.07)		a,b,c
MEANPUT	0.60 (0.25) (0.05, 2.15)	0.51 (0.28) (0.04, 2.17)	0.39 (0.16) (0.13, 0.69)		b
MEANCAU	1.58 (0.50) (0.28, 3.27)	1.30 (0.55) (0.17, 3.11)	1.04 (0.52) (0.24, 2.24)		a,b

Group	PD_CU	PD_MCI	PD_DEMENTIA	p-value	Post-Hoc
Subjects	323 (47.3%)	83 (12.2%)	17 (2.5%)		
Female	120 (37.2%)	20 (24.1%)	6 (35.3%)		None
APOE4	78 (26.4%) n=295	18 (23.7%) n=76	5 (38.5%) n=13		None
Age	60.1 (9.8) (33.5, 82.3)	66.0 (7.4) (48.0, 84.9)	69.4 (7.4) (54.8, 83.0)		a,b
Education	15.70 (2.87) (8.00, 26.00)	14.99 (2.82) (8.00, 21.00)	15.59 (4.91) (5.00, 21.00)		None
GDS	2.17 (2.40) (0.00, 14.00)	2.86 (2.61) (0.00, 13.00)	2.71 (2.11) (0.00, 8.00)		None
UPSIT	23.41 (7.87) (5.00, 39.00)	19.54 (8.60) (1.00, 40.00)	16.06 (7.39) (5.00, 31.00)		a,b
REM	3.82 (2.54) (0.00, 12.00)	5.06 (2.90) (0.00, 12.00)	5.24 (3.11) (0.00, 12.00)		a
UPDRS3	20.51 (8.73) (5.00, 47.00)	22.29 (9.40) (4.00, 51.00)	21.29 (8.18) (10.00, 39.00)		None
UPDRSTOT	31.35 (12.94) (7.00, 72.00)	35.35 (13.12) (7.00, 68.00)	36.94 (14.37) (18.00, 70.00)		None
UPDRSTOTOM	31.35 (12.94) (7.00, 72.00)	35.35 (13.12) (7.00, 68.00)	36.94 (14.37) (18.00, 70.00)		None
UPDRS1	5.21 (3.90) (0.00, 24.00)	6.59 (4.42) (0.00, 18.00)	7.53 (4.17) (2.00, 18.00)		
UPDRS2	5.63 (4.13) (0.00, 22.00)	6.54 (4.24) (1.00, 20.00)	8.12 (4.27) (2.00, 15.00)		None
UPDRS3ON	20.51 (8.73) (5.00, 47.00)	22.29 (9.40) (4.00, 51.00)	21.29 (8.18) (10.00, 39.00)		None
MOCA	27.33 (2.17) (19.00, 30.00)	26.58 (2.70) (17.00, 30.00)	25.94 (2.38) (21.00, 29.00)		
LNS	11.04 (2.50) (2.00, 20.00)	9.29 (2.58) (4.00, 17.00)	8.29 (2.78) (3.00, 13.00)		a,b
SDMT	42.94 (8.75) (17.00, 82.00)	36.69 (10.18) (7.00, 58.00)	29.82 (10.85) (15.00, 50.00)		a,b
JLO	12.98 (1.96) (5.00, 15.00)	12.19 (2.44) (6.00, 15.00)	11.53 (2.67) (6.00, 15.00)		a
INMEDIATE	25.26 (4.74) (9.00, 36.00)	22.13 (4.90) (11.00, 34.00)	20.24 (4.28) (14.00, 28.00)		a,b
HVLTRDLY	8.75 (2.35) (0.00, 12.00)	7.13 (2.75) (0.00, 12.00)	7.00 (2.12) (3.00, 11.00)		a
SFT	50.60 (11.54) (24.00, 103.00)	43.48 (9.39) (20.00, 70.00)	37.35 (9.37) (25.00, 54.00)		a,b
ADL	93.25 (6.02) (70.00, 100.00)	93.01 (5.29) (80.00, 100.00)	91.76 (6.11) (80.00, 100.00)		None
SCOPA	8.74 (5.70) (0.00, 35.00)	11.79 (6.96) (3.00, 39.00)	13.06 (7.20) (4.00, 32.00)		a
ABETA	942.40 (413.12) (315.90, 3707.00)	844.08 (395.26) (238.80, 2572.00)	592.27 (251.10) (308.80, 1300.00)		b
ASYN	1519.17 (680.58) (432.40, 5256.90)	1470.97 (620.62) (472.00, 3443.70)	1443.49 (642.42) (539.40, 2780.50)		None
TAU	167.17 (54.61) (80.93, 467.00)	172.53 (59.27) (83.93, 345.30)	202.68 (78.07) (97.17, 327.40)		None
PTAU	14.52 (4.99) (8.02, 40.13)	15.51 (5.40) (8.36, 32.80)	18.59 (7.68) (8.01, 30.84)		b
pTauAB	0.02 (0.01) (0.01, 0.07)	0.02 (0.01) (0.01, 0.08)	0.03 (0.02) (0.01, 0.07)		b,c
MEANPUT	0.85 (0.29) (0.24, 2.15)	0.77 (0.32) (0.29, 2.17)	0.66 (0.18) (0.31, 1.05)		
MEANCAU	2.07 (0.54) (0.39, 3.71)	1.80 (0.54) (0.62, 3.27)	1.59 (0.50) (0.74, 2.52)		a,b

Group	PD_CU	PD_MCI	PD_DEMENTIA	p-value	Post-Hoc
Subjects	323 (47.3%)	83 (12.2%)	17 (2.5%)		
Female	120 (37.2%)	20 (24.1%)	6 (35.3%)	0.083	None
APOE4	78 (26.4%) n=295	18 (23.7%) n=76	5 (38.5%) n=13	0.101	None
Age	60.1 (9.8) (33.5, 82.3)	66.0 (7.4) (48.0, 84.9)	69.4 (7.4) (54.8, 83.0)	< 0.001	a,b
Education	15.70 (2.87) (8.00, 26.00)	14.99 (2.82) (8.00, 21.00)	15.59 (4.91) (5.00, 21.00)	0.150	None
GDS	2.17 (2.40) (0.00, 14.00)	2.86 (2.61) (0.00, 13.00)	2.71 (2.11) (0.00, 8.00)	0.058	None
UPSIT	23.41 (7.87) (5.00, 39.00)	19.54 (8.60) (1.00, 40.00)	16.06 (7.39) (5.00, 31.00)	< 0.001	a,b
REM	3.82 (2.54) (0.00, 12.00)	5.06 (2.90) (0.00, 12.00)	5.24 (3.11) (0.00, 12.00)	< 0.001	a
UPDRS3	20.51 (8.73) (5.00, 47.00)	22.29 (9.40) (4.00, 51.00)	21.29 (8.18) (10.00, 39.00)	0.260	None
MOCA	27.33 (2.17) (19.00, 30.00)	26.58 (2.70) (17.00, 30.00)	25.94 (2.38) (21.00, 29.00)	0.003	
LNS	11.04 (2.50) (2.00, 20.00)	9.29 (2.58) (4.00, 17.00)	8.29 (2.78) (3.00, 13.00)	< 0.001	a,b
SDMT	42.94 (8.75) (17.00, 82.00)	36.69 (10.18) (7.00, 58.00)	29.82 (10.85) (15.00, 50.00)	< 0.001	a,b
JLO	12.98 (1.96) (5.00, 15.00)	12.19 (2.44) (6.00, 15.00)	11.53 (2.67) (6.00, 15.00)	< 0.001	a
INMEDIATE	25.26 (4.74) (9.00, 36.00)	22.13 (4.90) (11.00, 34.00)	20.24 (4.28) (14.00, 28.00)	< 0.001	a,b
HVLTRDLY	8.75 (2.35) (0.00, 12.00)	7.13 (2.75) (0.00, 12.00)	7.00 (2.12) (3.00, 11.00)	< 0.001	a
SFT	50.60 (11.54) (24.00, 103.00)	43.48 (9.39) (20.00, 70.00)	37.35 (9.37) (25.00, 54.00)	< 0.001	a,b
ADL	93.25 (6.02) (70.00, 100.00)	93.01 (5.29) (80.00, 100.00)	91.76 (6.11) (80.00, 100.00)	0.583	None
SCOPA	8.74 (5.70) (0.00, 35.00)	11.79 (6.96) (3.00, 39.00)	13.06 (7.20) (4.00, 32.00)	< 0.001	a
ABETA	942.40 (413.12) (315.90, 3707.00)	844.08 (395.26) (238.80, 2572.00)	592.27 (251.10) (308.80, 1300.00)	0.001	b
ASYN	1519.17 (680.58) (432.40, 5256.90)	1470.97 (620.62) (472.00, 3443.70)	1443.49 (642.42) (539.40, 2780.50)	0.783	None
TAU	167.17 (54.61) (80.93, 467.00)	172.53 (59.27) (83.93, 345.30)	202.68 (78.07) (97.17, 327.40)	0.053	None
PTAU	14.52 (4.99) (8.02, 40.13)	15.51 (5.40) (8.36, 32.80)	18.59 (7.68) (8.01, 30.84)	0.007	b
pTauAB	0.0161 (0.0071) (0.0063, 0.0673)	0.0196 (0.0120) (0.0103, 0.0816)	0.0337 (0.0169) (0.0138, 0.0698)	< 0.001	b,c
MEANPUT	0.85 (0.29) (0.24, 2.15)	0.77 (0.32) (0.29, 2.17)	0.66 (0.18) (0.31, 1.05)	0.005	
MEANCAU	2.07 (0.54) (0.39, 3.71)	1.80 (0.54) (0.62, 3.27)	1.59 (0.50) (0.74, 2.52)	< 0.001	a,b